A065-v0.4.0 | A003 volume and numerical Euler campaign

Result: EXECUTED

Scope: periodic 3D Euler IVP from an ASSUMED Gaussian filament initialization.

Not a source-native PKLSA Euler provider; no independent physical pressure holdout.

Precommit SHA-256: f63242516db654c2ecfeb2d45744d68c7438ca5c7fa48cdf0ec1822a8afa9437

Gates

G37 PASS

G38 PASS

G39 PASS

G40 PASS

G41 UNRESOLVED

G42 PASS

Independent solver numerical control: Taylor-Green

Grid 24 T = 0.02 dt = 0.005 RK4 steps 4

Relative energy drift = 1.146e-16

div(u) RMS = 3.063e-18

RK4 dt-halving relative u difference = 1.153e-13

Control outcome = PASS

Source: torus_3_1_direct

Verified A003 XYZ SHA-256 16f34a900af097e893a0ad15c3f0b8d0c35d74dbd6edac11d8d60057a98f16e3

Fixed: Gamma=1; Gaussian a=0.55; L=12; 200 source nodes; T=0.02

N=48: admitted sections=3; Euler energy drift 2.3e-16; divergence 1.91e-18

N=64: admitted sections=3; Euler energy drift 2.3e-16; divergence 1.88e-18

N=96: admitted sections=3; Euler energy drift 0; divergence 1.86e-18

48 -> 64, source omega 0.000%, u 0.000%, p 0.002%

Euler omega 0.227%, u 0.037%, p 0.020%

64 -> 96, source omega 0.000%, u 0.000%, p 0.000%

Euler omega 0.002%, u 0.000%, p 0.000%

VOLUME True; EULER NUMERICS True; PRESSURE GRID True

Source: torus_3_1_hoste

Verified A003 XYZ SHA-256 123cd798fc754269ac2baf4225b4c8097c2e47cb5d46b9e814e7f2d74712906d

Fixed: Gamma=1; Gaussian a=0.55; L=12; 200 source nodes; T=0.02

N=48: admitted sections=1; Euler energy drift 6.26e-16; divergence 1.18e-18

N=64: admitted sections=1; Euler energy drift 8.35e-16; divergence 1.15e-18

N=96: admitted sections=1; Euler energy drift 6.26e-16; divergence 1.19e-18

48 -> 64, source omega 0.000%, u 0.000%, p 0.002%

Euler omega 0.236%, u 0.053%, p 0.024%

64 -> 96, source omega 0.000%, u 0.000%, p 0.000%

Euler omega 0.002%, u 0.000%, p 0.000%

VOLUME True; EULER NUMERICS True; PRESSURE GRID True

Interpretation

G38 and G42 test discretization consistency, not the correctness of Gaussian core physics.

G39 and G40 qualify Euler numerical time stepping, not the physical initial condition.

G41 remains UNRESOLVED until a separately provenance-certified PKLSA Euler source

and an independently derived/measured physical velocity/pressure field are provided.